Tarea 3

Patrones de Diseñño

> *Programación Avanzada SC-701 M 6-9 Fecha de Entrega: Miércoles 19 de Noviembre Trabajo Individual (4%)*

Entregables

> *Documento en formato pdf | estándar IEEE paper format o Apa7 Archivo de entrega: Nombre_Apellido_M_T3.zip*

Indicaciones

1. Realizar todos los puntos.
2. El proyecto TIENE que compilar de lo contrario no tiene los puntos
3. El proyecto sera probado con codigo (Unit Testing)

Parte 1

- Modificar la BD con el nuevo script (Sugerencia: elimine todas las tablas y corra el script nuevamente)
- Crear una solucion nueva
- Corra el scaffolding en el proyecto de PAW.Data
- Elimine las entidades existentes en PAW.Models
- Mueva las entidades generadas al proyecto de PAW.Models
  - Cambie el namespace de esas nuevas entidades a `namespace PAW.Models;`
- Compile el codigo y asegurese de que no tenga errores

Parte 2

- Puede reutilizar el proyecto de PAW.Architecture, PAW.Repositories y PAW.Service en la nueva solucion
  - Busque como agregar un proyecto existente (google :-))
- Agregar los repositorios de las clases:
  - Categories, Products, Inventory, Roles, User, UserRoles y Task
- Crear los managers de esas clases (todas)
- Crear los API Cntrollers de esas clases (menos tasks) con un crud basido mas el GetById
  - ReadAllAsync, GetByIdAsync, DeleteAsync, UpdateAsync, CreateAsync
- Al final los metodos deben verse en el menu de SwaggerUI al levantar la aplicacion
- TODOS los metodos deben ser `async Tas<>`
- Patrones a utilizar: Dependency Injection/IoC, Repository, Base, Service

Parte 3

- Crear un minimal API con la clase Task
  - Debe crearlo en un empty API project
  - Todos los metodos deben funcionar y tener la siguiente estructura:
    - GET /tasks
    - GET /tasks/{id}
    - POST /tasks
    - PUT / tasks/{id}

- DELETE /tasks{id}

Script de la base de datos

No modificar la estructura

```
USE [ProductDB]
GO
/****** Object:  Table [dbo].[Categories]    Script Date: 2/12/2025 11:03:19 PM
******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Categories](
    [CategoryID] [int] IDENTITY(1,1) NOT NULL,
    [CategoryName] [nvarchar](255) NULL,
    [Description] [nvarchar](1000) NULL,
    [LastModified] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [CategoryID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Inventory]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Inventory](
    [InventoryID] [int] IDENTITY(1,1) NOT NULL,
    [UnitPrice] [decimal](10, 2) NULL,
    [UnitsInStock] [int] NULL,
    [LastUpdated] [datetime] NULL,
    [ProductId] [int] NULL,
    [DateAdded] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [InventoryID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Products]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
```

```
GO
CREATE TABLE [dbo].[Products](
    [ProductID] [int] IDENTITY(1,1) NOT NULL,
    [ProductName] [nvarchar](255) NULL,
    [InventoryID] [int] NULL,
    [SupplierID] [int] NULL,
    [Description] [nvarchar](1000) NULL,
    [Rating] [decimal](3, 2) NULL,
    [CategoryID] [int] NULL,
    [LastModified] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [ProductID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Roles]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Roles](
    [RoleID] [int] IDENTITY(1,1) NOT NULL,
    [RoleName] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [RoleID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Suppliers]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Suppliers](
    [SupplierID] [int] IDENTITY(1,1) NOT NULL,
    [SupplierName] [nvarchar](255) NULL,
    [ContactName] [nvarchar](255) NULL,
    [ContactTitle] [nvarchar](255) NULL,
    [Phone] [nvarchar](255) NULL,
    [Address] [nvarchar](1000) NULL,
    [City] [nvarchar](255) NULL,
    [Country] [nvarchar](255) NULL,
    [LastModified] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
```

```
(
    [SupplierID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Tasks]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Tasks](
    [Id] [int] IDENTITY(1,1) NOT NULL,
    [Name] [nvarchar](255) NULL,
    [Description] [nvarchar](1000) NULL,
    [Status] [nvarchar](50) NULL,
    [DueDate] [datetime] NULL,
    [CreatedAt] [datetime] NULL,
    [LastModified] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [Id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
/****** Object:  Table [dbo].[Users]    Script Date: 2/12/2025 11:03:19 PM ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Users](
    [UserID] [int] IDENTITY(1,1) NOT NULL,
    [Username] [nvarchar](255) NULL,
    [Email] [nvarchar](255) NULL,
    [PasswordHash] [nvarchar](255) NULL,
    [CreatedAt] [datetime] NULL,
    [IsActive] [bit] NULL,
    [LastModified] [datetime] NULL,
    [ModifiedBy] [nvarchar](255) NULL,
PRIMARY KEY CLUSTERED
(
    [UserID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
SET IDENTITY_INSERT [dbo].[Categories] ON
```

```
GO
INSERT [dbo].[Categories] ([CategoryID], [CategoryName], [Description],
[LastModified], [ModifiedBy]) VALUES (1, N'Beverages', N'Drinks like soda, water, and
juice', CAST(N'2025-02-12T20:44:38.100' AS DateTime), N'System')
GO
INSERT [dbo].[Categories] ([CategoryID], [CategoryName], [Description],
[LastModified], [ModifiedBy]) VALUES (2, N'Food', N'All kinds of food items',
CAST(N'2025-02-12T20:44:38.100' AS DateTime), N'System')
GO
INSERT [dbo].[Categories] ([CategoryID], [CategoryName], [Description],
[LastModified], [ModifiedBy]) VALUES (3, N'Snacks', N'Snack items like chips and
candy', CAST(N'2025-02-12T20:44:38.100' AS DateTime), N'System')
GO
INSERT [dbo].[Categories] ([CategoryID], [CategoryName], [Description],
[LastModified], [ModifiedBy]) VALUES (4, N'Dairy', N'Milk, cheese, and other dairy
products', CAST(N'2025-02-12T20:44:38.100' AS DateTime), N'System')
GO
INSERT [dbo].[Categories] ([CategoryID], [CategoryName], [Description],
[LastModified], [ModifiedBy]) VALUES (5, N'Electronics', N'Devices and gadgets',
CAST(N'2025-02-12T20:44:38.100' AS DateTime), N'System')
GO
SET IDENTITY_INSERT [dbo].[Categories] OFF
GO
SET IDENTITY_INSERT [dbo].[Inventory] ON
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (1, CAST(2.99 AS Decimal(10, 2)), 200,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 1, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (2, CAST(3.49 AS Decimal(10, 2)), 150,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 2, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (3, CAST(5.99 AS Decimal(10, 2)), 100,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 3, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (4, CAST(1.99 AS Decimal(10, 2)), 250,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 4, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (5, CAST(2.49 AS Decimal(10, 2)), 180,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 5, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (6, CAST(3.49 AS Decimal(10, 2)), 200,
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CAST(N'2025-02-12T20:44:48.367' AS DateTime), 6, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (7, CAST(199.99 AS Decimal(10, 2)), 50,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 7, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (8, CAST(999.99 AS Decimal(10, 2)), 30,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 8, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (9, CAST(799.99 AS Decimal(10, 2)), 40,
CAST(N'2025-02-12T20:44:48.367' AS DateTime), 9, CAST(N'2025-02-12T20:44:48.367' AS
DateTime), N'System')
GO
INSERT [dbo].[Inventory] ([InventoryID], [UnitPrice], [UnitsInStock], [LastUpdated],
[ProductId], [DateAdded], [ModifiedBy]) VALUES (10, CAST(149.99 AS Decimal(10, 2)),
60, CAST(N'2025-02-12T20:44:48.367' AS DateTime), 10, CAST(N'2025-02-12T20:44:48.367'
AS DateTime), N'System')
GO
SET IDENTITY_INSERT [dbo].[Inventory] OFF
GO
SET IDENTITY_INSERT [dbo].[Products] ON
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID],
[Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (2,
N'Coca-Cola', 1, 1, N'Refreshing soda drink', CAST(4.50 AS Decimal(3, 2)), 1,
CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID],
[Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (3,
N'Pepsi', 2, 2, N'Popular soda drink', CAST(4.30 AS Decimal(3, 2)), 1, CAST(N'2025-02-
12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID],
[Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (4, N'Oreo
Cookies', 3, 3, N'Chocolate sandwich cookies', CAST(4.70 AS Decimal(3, 2)), 3,
CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID],
[Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (5, N'Lays
Chips', 4, 4, N'Crunchy potato chips', CAST(4.40 AS Decimal(3, 2)), 3, CAST(N'2025-02-
12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID],
[Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (6,
N'Cheetos', 5, 5, N'Spicy cheesy snacks', CAST(4.60 AS Decimal(3, 2)), 3, CAST(N'2025-
02-12T20:45:12.400' AS DateTime), N'System')
GO
```

```
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID], [Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (7, N'Nestle Milk Chocolate', 6, 6, N'Milk chocolate bar', CAST(4.80 AS Decimal(3, 2)), 3, CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID], [Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (8, N'Samsung TV', 7, 7, N'High-definition TV', CAST(4.50 AS Decimal(3, 2)), 5, CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID], [Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (9, N'iPhone 13', 8, 8, N'Latest model of iPhone', CAST(4.90 AS Decimal(3, 2)), 5, CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID], [Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (10, N'Dell Laptop', 9, 9, N'Powerful laptop for work and play', CAST(4.70 AS Decimal(3, 2)), 5, CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
INSERT [dbo].[Products] ([ProductID], [ProductName], [InventoryID], [SupplierID], [Description], [Rating], [CategoryID], [LastModified], [ModifiedBy]) VALUES (11, N'Smart Watch', 10, 10, N'Wearable fitness tracker', CAST(4.40 AS Decimal(3, 2)), 5, CAST(N'2025-02-12T20:45:12.400' AS DateTime), N'System')
GO
SET IDENTITY_INSERT [dbo].[Products] OFF
GO
SET IDENTITY_INSERT [dbo].[Roles] ON
GO
INSERT [dbo].[Roles] ([RoleID], [RoleName]) VALUES (1, N'Admin')
GO
INSERT [dbo].[Roles] ([RoleID], [RoleName]) VALUES (2, N'User')
GO
INSERT [dbo].[Roles] ([RoleID], [RoleName]) VALUES (3, N'Manager')
GO
INSERT [dbo].[Roles] ([RoleID], [RoleName]) VALUES (4, N'Support')
GO
INSERT [dbo].[Roles] ([RoleID], [RoleName]) VALUES (5, N'Guest')
GO
SET IDENTITY_INSERT [dbo].[Roles] OFF
GO
SET IDENTITY_INSERT [dbo].[Suppliers] ON
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (1, N'Acme Corp', N'John Smith', N'Sales Manager', N'123-456-7890', N'123 Acme St.', N'New York', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (2, N'Global Supplies', N'Jane Doe', N'CEO', N'987-654-3210', N'456 Global Ave.', N'Los Angeles', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
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GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (3, N'Tech Innovators', N'Mike Johnson', N'CTO', N'555-123-4567', N'789 Tech Rd.', N'San Francisco', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (4, N'Prime Goods', N'Alice White', N'Marketing Lead', N'321-654-9870', N'321 Prime Blvd.', N'Chicago', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (5, N'Best Products', N'Bob Green', N'Logistics Manager', N'876-543-2109', N'654 Best St.', N'Houston', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (6, N'Quick Supplies', N'Lily Adams', N'Sales Director', N'123-789-4560', N'159 Quick Ln.', N'Seattle', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (7, N'Fresh Foods', N'Eve Black', N'Operations Lead', N'234-567-8901', N'753 Fresh Ave.', N'Miami', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (8, N'Super Electronics', N'Jack Hill', N'Product Manager', N'345-678-9012', N'321 Super Dr.', N'Austin', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (9, N'Luxury Living', N'Rachel Scott', N'CEO', N'456-789-0123', N'432 Luxury Rd.', N'Dallas', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
INSERT [dbo].[Suppliers] ([SupplierID], [SupplierName], [ContactName], [ContactTitle], [Phone], [Address], [City], [Country], [LastModified], [ModifiedBy]) VALUES (10, N'Tech World', N'Tom Lewis', N'COO', N'567-890-1234', N'543 Tech Ln.', N'Boston', N'USA', CAST(N'2025-02-12T20:45:05.617' AS DateTime), N'System')
GO
SET IDENTITY_INSERT [dbo].[Suppliers] OFF
GO
SET IDENTITY_INSERT [dbo].[Tasks] ON
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (1, N'Fix bug #123', N'Resolve the issue with the payment gateway', N'In Progress', CAST(N'2025-02-14T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt],
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[LastModified], [ModifiedBy]) VALUES (2, N'Update website', N'Redesign the home page and improve UX', N'Pending', CAST(N'2025-02-17T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (3, N'Deploy new release', N'Deploy the new feature release to production', N'Completed', CAST(N'2025-02-15T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (4, N'Customer feedback', N'Analyze the customer survey results', N'Pending', CAST(N'2025-02-19T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (5, N'Update inventory', N'Add new items to the product list', N'In Progress', CAST(N'2025-02-13T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (6, N'Send newsletter', N'Prepare and send out the weekly newsletter', N'Completed', CAST(N'2025-02-18T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (7, N'Audit logs', N'Check system activity logs for issues', N'In Progress', CAST(N'2025-02-15T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (8, N'Fix broken links', N'Resolve broken URLs on the website', N'Pending', CAST(N'2025-02-14T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (9, N'Prepare demo', N'Prepare a demo of the new feature for the client', N'Completed', CAST(N'2025-02-16T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
GO
INSERT [dbo].[Tasks] ([Id], [Name], [Description], [Status], [DueDate], [CreatedAt], [LastModified], [ModifiedBy]) VALUES (10, N'Performance optimization', N'Optimize server performance and response times', N'In Progress', CAST(N'2025-02-17T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), CAST(N'2025-02-12T20:45:01.647' AS DateTime), N'System')
```

```
GO
SET IDENTITY_INSERT [dbo].[Tasks] OFF
GO
SET IDENTITY_INSERT [dbo].[Users] ON
GO
INSERT [dbo].[Users] ([UserID], [Username], [Email], [PasswordHash], [CreatedAt],
[IsActive], [LastModified], [ModifiedBy]) VALUES (1, N'john_doe',
N'john.doe@example.com', N'hashedpassword1', CAST(N'2025-02-12T20:44:56.373' AS
DateTime), 1, CAST(N'2025-02-12T20:44:56.373' AS DateTime), N'System')
GO
INSERT [dbo].[Users] ([UserID], [Username], [Email], [PasswordHash], [CreatedAt],
[IsActive], [LastModified], [ModifiedBy]) VALUES (2, N'jane_smith',
N'jane.smith@example.com', N'hashedpassword2', CAST(N'2025-02-12T20:44:56.373' AS
DateTime), 1, CAST(N'2025-02-12T20:44:56.373' AS DateTime), N'System')
GO
INSERT [dbo].[Users] ([UserID], [Username], [Email], [PasswordHash], [CreatedAt],
[IsActive], [LastModified], [ModifiedBy]) VALUES (3, N'mike_jones',
N'mike.jones@example.com', N'hashedpassword3', CAST(N'2025-02-12T20:44:56.373' AS
DateTime), 1, CAST(N'2025-02-12T20:44:56.373' AS DateTime), N'System')
GO
INSERT [dbo].[Users] ([UserID], [Username], [Email], [PasswordHash], [CreatedAt],
[IsActive], [LastModified], [ModifiedBy]) VALUES (4, N'alice_brown',
N'alice.brown@example.com', N'hashedpassword4', CAST(N'2025-02-12T20:44:56.373' AS
DateTime), 1, CAST(N'2025-02-12T20:44:56.373' AS DateTime), N'System')
GO
INSERT [dbo].[Users] ([UserID], [Username], [Email], [PasswordHash], [CreatedAt],
[IsActive], [LastModified], [ModifiedBy]) VALUES (5, N'bob_white',
N'bob.white@example.com', N'hashedpassword5', CAST(N'2025-02-12T20:44:56.373' AS
DateTime), 1, CAST(N'2025-02-12T20:44:56.373' AS DateTime), N'System')
GO
SET IDENTITY_INSERT [dbo].[Users] OFF
GO
ALTER TABLE [dbo].[Products]  WITH CHECK ADD FOREIGN KEY([CategoryID])
REFERENCES [dbo].[Categories] ([CategoryID])
GO
ALTER TABLE [dbo].[Products]  WITH CHECK ADD FOREIGN KEY([InventoryID])
REFERENCES [dbo].[Inventory] ([InventoryID])
GO
ALTER TABLE [dbo].[Products]  WITH CHECK ADD FOREIGN KEY([SupplierID])
REFERENCES [dbo].[Suppliers] ([SupplierID])
GO
```

Calificación

Tarea	Descripción	Peso
Parte 1		
Modificar la BD con el nuevo script (eliminar todas las tablas y volver a ejecutar el script)	Recrear el esquema de la base de datos con el script proporcionado.	10
Crear una solución nueva	Configurar una nueva solución en el	5

	proyecto.	
Ejecutar el scaffolding en el proyecto PAW.Data	Generar el código scaffolding para crear las entidades y DbContext.	5
Eliminar las entidades existentes en PAW.Models	Eliminar las entidades anteriores en el proyecto PAW.Models.	5
Mover las entidades generadas al proyecto PAW.Models	Transferir las entidades generadas desde PAW.Data a PAW.Models y cambiar el namespace.	5
Compilar el código y asegurarse de que no tenga errores	Asegurarse de que la solución compile correctamente sin errores.	5
Parte 2		
Reutilizar los proyectos PAW.Architecture, PAW.Repositories y PAW.Service en la nueva solución	Agregar estos proyectos existentes a la nueva solución.	5
Agregar los repositorios para las clases: Categories, Products, Inventory, Roles, User, UserRoles y Task	Implementar repositorios para cada una de estas entidades.	10
Crear los managers para las clases mencionadas	Implementar los managers (capa de lógica de negocio) para cada repositorio.	10
Crear los API Controllers para las clases (excepto Task) con operaciones CRUD y GetById	Implementar las operaciones CRUD para todas las entidades excepto Task, incluyendo métodos asincrónicos.	10
Asegurarse de que los métodos aparezcan en SwaggerUI al ejecutar la aplicación	Probar la API para asegurarse de que SwaggerUI muestre los métodos de la API correctamente.	5
Todos los métodos deben ser asincrónicos (`async Task<>`)	Implementar métodos asincrónicos para todas las operaciones CRUD.	5
Utilizar los patrones adecuados: Dependency Injection/IoC, Repository, Base, Service	Asegurarse de que el diseño siga los patrones recomendados de arquitectura de software.	10
Parte 3		
Crear una minimal API con la clase Task en un proyecto API vacío	Configurar un proyecto minimal API con la clase Task.	10
Asegurarse de que todos los métodos funcionen para Task (GET, POST, PUT, DELETE)	Implementar correctamente los métodos HTTP para Task. Asegurarse de que funcionen como se espera.	10
Asegurarse de que los siguientes endpoints funcionen:		

- GET /tasks	Endpoint para obtener todas las tareas.	5
- GET /tasks/{id}	Endpoint para obtener una tarea por ID.	5
- POST /tasks	Endpoint para crear una nueva tarea.	5
- PUT /tasks/{id}	Endpoint para actualizar una tarea existente.	5
- DELETE /tasks/{id}	Endpoint para eliminar una tarea.	5
Total		**100**